

Rust Systems Programming & Architecture

Industry-Standard Profiling, Coverage & Benchmarking Tools

Compiled by Antigravity AI • DeepMind Advanced Agentic Coding

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Guide: Statistical Micro-Benchmarking with Criterion.rs

Overview

Criterion.rs is the industry standard for statistics-driven micro-benchmarking in Rust. Inspired by Haskell's Criterion library, it runs benchmarks across thousands of iterations, performs rigorous statistical regression analysis (e.g., detecting a +1.2% latency increase with 95% confidence), and generates visual HTML charts using TinyHTML while isolating compiler optimization noise.

1. How to Download & Install (download)

Criterion is configured as a development dependency in your project's `Cargo.toml`.

Step 1: Add Dependency & Disable Default Harness

Add `criterion` under `[dev-dependencies]`, and declare a `[[bench]]` target with `harness = false` (which tells Cargo to use Criterion's custom benchmark runner instead of the standard library's test harness):

```
[dev-dependencies]
criterion = { version = "0.5", features = ["html_reports"] }

[[bench]]
name = "my_benchmark"
harness = false
```

Step 2: Create the Benchmark File

Create a new directory named `benches/` at the root of your project, and create `benches/my_benchmark.rs`.

2. How to Configure for Debugging (debug)

To write accurate micro-benchmarks, you must prevent LLVM from performing **Dead Code Elimination (DCE)**. If you compute a value inside a benchmark loop but never use it, LLVM's optimizer will delete your entire algorithm at compile time, resulting in fake `0.000 ns` benchmark times!

The `black_box` Solution

Always wrap your benchmark inputs and outputs in `std::hint::black_box(...)` (or `criterion::black_box(...)`). This forces the compiler to assume the value is dynamically read and written to opaque memory.

Runnable Benchmark Template (`benches/my_benchmark.rs`):

```
use criterion::{black_box, criterion_group, criterion_main, Criterion};

// The function we want to benchmark
fn fibonacci(n: u64) -> u64 {
    match n {
        0 => 1,
        1 => 1,
        _ => fibonacci(n - 1) + fibonacci(n - 2),
    }
}

fn bench_fibonacci(c: &mut Criterion) {
    // 1. Define a benchmark group or single routine
    c.bench_function("fib 20", |b| {
        // 2. b.iter(...) runs the closure thousands of times to gather statistics
        b.iter(|| {
            // 3. Wrap both input and output in black_box!
            fibonacci(black_box(20))
        })
    });
}

// Register benchmark functions and generate main()
criterion_group!(benches, bench_fibonacci);
criterion_main!(benches);
```

3. How to Run & Profile Benchmarks (`profile`)

Run All Benchmarks

To execute your benchmarks and perform statistical analysis against previous runs:

```
cargo bench
```

Run a Specific Benchmark Filter

If you have multiple benchmark suites:

```
cargo bench -- "fib 20"
```

Save a Baseline for Regression Testing

When making optimization experiments, save your current performance as a named baseline:

```
# Save current code performance as "before_pr":
cargo bench -- --save-baseline before_pr

# Make your code edits... then compare against the baseline:
cargo bench -- --baseline before_pr
```

4. Interpreting Results & Best Practices

When `cargo bench` finishes, it outputs statistical summaries to the terminal and generates an interactive HTML report suite under `target/criterion/report/index.html`.

Terminal Output Breakdown:

```
fib 20                time:   [26.142 µs 26.205 µs 26.273 µs]
                      change: [-3.412% -2.105% -0.891%] (p = 0.00 < 0.05)
                      Performance has improved.
```

- **time: [Low Median High]**: The 95% confidence interval of execution time per iteration.
- **change: [Low Median High]**: The percentage difference compared to the last recorded baseline!
- **p = 0.00 < 0.05**: The statistical p-value confirming whether the change is a genuine performance shift or random system noise.

Best Practices:

- **Close Background Apps**: Close web browsers, Docker containers, and Slack before running benchmarks to prevent CPU throttling and OS scheduling noise.
- **Benchmark Across Input Sizes**: Use `c.bench_with_input(...)` and `BenchmarkId` to test algorithms across vectors of size 10, 1,000, and 100,000 elements to observe asymptotic scaling ($O(N)$ vs $O(N \log N)$).

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Guide: Heap Memory Profiling & Leak Detection with DHAT

Overview

DHAT (Dynamic Heap Analysis Tool) is an industry-standard heap profiler for Rust. Built as a native Rust implementation of Valgrind's DHAT algorithm, it intercepts every heap allocation (`Box`, `Vec`, `String`, `HashMap`, etc.) to track allocation counts, peak memory consumption, short-lived heap churn, and memory leaks at program termination.

1. How to Download & Install (`download`)

Unlike external command-line tools, `dhat` is integrated directly into your Rust application as a library crate and custom global allocator.

Add `dhat` to your project's `Cargo.toml`:

```
[dependencies]
dhat = "0.3"
```

2. How to Configure for Debugging (`debug`)

To allow DHAT to record exact file names and line numbers for every allocation and memory leak, ensure debug symbols are enabled in `Cargo.toml`:

```
[profile.release]
debug = true           # Required for line-by-line allocation stack traces!
strip = false
```

Setting up the Global Allocator in Code

In your `src/main.rs` (or test harness), register DHAT as the program's global allocator and initialize the profiler:

```

use dhat::{Dhat, Profiler};

// 1. Register DHAT as the global heap allocator
#[global_allocator]
static ALLOC: Dhat = Dhat;

fn main() {
    // 2. Start the heap profiler as the very first line in main()
    let _profiler = Profiler::new_heap();

    println!("Starting application execution...");

    // Simulate some heap work and an intentional memory leak!
    let mut numbers = Vec::with_capacity(10_000);
    numbers.extend(0..10_000);

    // Leaking memory intentionally to demonstrate leak detection:
    let _leaked_box = Box::leak(Box::new(vec!["leak", "memory", "forever"]));

    println!("Application finishing...");
    // 3. When `_profiler` is dropped at the end of main(), DHAT outputs
    // its summary to stderr and writes `dhat-heap.json` to disk!
}

```

3. How to Run & Profile (`profile`)

Compile and run your application in release mode:

```
cargo run --release
```

Terminal Output Summary

When execution finishes, DHAT prints a high-level summary to `stderr`:

```

dhat: Total:      80,072 bytes in 2 blocks
dhat: At t-gmax: 80,000 bytes in 1 blocks
dhat: At t-end:   72 bytes in 1 blocks (LEAKED!)
dhat: The data has been saved to dhat-heap.json, and is viewable with dhat/dh-view.html

```

- **Total** : The cumulative sum of all heap allocations across the entire program lifespan.
- **At t-gmax** : The exact moment of **Peak Heap Memory Usage** (global maximum).
- **At t-end** : Memory that was **never deallocated** when the program ended (**Memory Leaks!**).

4. Interpreting Results & Best Practices

To view the detailed interactive report, open the generated `dhat-heap.json` file using the official online DHAT viewer:

1. Open https://nnetherote.github.io/dh_view/dh_view.html in your web browser.
2. Click "Load..." and select your local `dhat-heap.json` file.

Understanding the DHAT Viewer:

- **Sort by `Total` (Heap Churn):** Shows functions creating millions of short-lived temporary objects (e.g., allocating intermediate `String`s inside a loop instead of reusing a buffer).
- **Sort by `t-gmax` (Peak Memory):** Shows which data structures were sitting in RAM when your application hit its highest memory consumption. Great for sizing container capacities!
- **Sort by `t-end` (Memory Leaks):** Shows exact stack traces of memory that was leaked via `Box::Leak`, `std::mem::forget`, or reference-counted cycles (`Rc` / `Arc`).

Best Practices:

- Remove or conditionally compile `#[global_allocator] static ALLOC: Dhat = Dhat;` in production builds, as DHAT adds runtime tracking overhead to every malloc/free call.
- Use `#[cfg(feature = "dhat-heap")]` to toggle profiling cleanly via `cargo run --release --features dhat-heap`.

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Guide: Fast & Lightweight Micro-Benchmarking with Divan

Overview

Divan is a modern, ultra-fast micro-benchmarking framework for Rust. Designed as a simpler and significantly faster alternative to Criterion.rs, Divan leverages Rust attribute macros (`#[divan::bench]`), requires zero external HTML dependencies, natively counts heap allocations per benchmark, and supports effortless generic type benchmarking across multiple input sizes.

1. How to Download & Install (`download`)

Add `divan` as a development dependency in your `Cargo.toml`, and declare your benchmark target with `harness = false`:

```
[dev-dependencies]
divan = "0.1"

[[bench]]
name = "my_benchmark"
harness = false
```

Create your benchmark file under `benches/my_benchmark.rs`.

2. How to Configure for Debugging (`debug`)

Divan makes writing benchmark harnesses incredibly concise using procedural attribute macros. Like Criterion, always wrap inputs/outputs in `divan::black_box` to prevent compiler dead-code elimination.

Runnable Benchmark Template (`benches/my_benchmark.rs`):

```
use divan::{black_box, Bencher};

fn main() {
    // Run all registered Divan benchmarks
    divan::main();
}

// 1. Simple benchmark using attribute macro
#[divan::bench]
fn bench_sorting() {
    let mut data = vec![5, 2, 8, 1, 9, 3];
    data.sort();
    black_box(data);
}

// 2. Benchmarking across multiple generic types!
#[divan::bench(types = [i32, i64, f64])]
fn bench_math<T: Copy + std::ops::Add<Output = T> + Default>() {
    let a = black_box(T::default());
    let b = black_box(T::default());
    black_box(a + b);
}

// 3. Benchmarking across multiple input sample sizes!
#[divan::bench(args = [10, 1_000, 100_000])]
fn bench_vec_allocation(len: usize) {
    let vec: Vec<u64> = Vec::with_capacity(black_box(len));
    black_box(vec);
}
```

3. How to Run & Profile Benchmarks (`profile`)

Run All Benchmarks

To execute your benchmarks and view the terminal histogram:

```
cargo bench
```

Enable Heap Allocation Tracking

One of Divan's greatest superpowers is its ability to track exact heap allocation counts and bytes allocated per iteration without needing external tools like DHAT!

```
cargo bench -- --alloc
```

Filter Benchmarks by Name

```
cargo bench -- bench_vec_allocation
```

4. Interpreting Results & Best Practices

Divan outputs clean, colorful histograms and comparative tables directly to your terminal without cluttering your disk with HTML files.

Terminal Output Breakdown (with `--alloc`):

```
Timer precision: 41 ns
my_benchmark          fastest      | slowest      | median      | mean      | samples
├─ iter | alloc / iter
├─ bench_vec_allocation
│   └─ 10
│       800 | 1 B, 1 count      14.21 ns      | 45.12 ns      | 15.01 ns      | 16.32 ns      | 10000
│   └─ 1000
│       800 | 8 KB, 1 count    112.4 ns      | 310.2 ns      | 118.5 ns      | 122.1 ns      | 10000
│   └─ 100000
│       800 | 800 KB, 1 count 8.412 µs      | 24.15 µs      | 8.910 µs      | 9.102 µs      | 10000
```

- `median` / `mean`: The central tendency of execution latency.
- `alloc / iter`: Shows exactly how many heap allocations (`1 count`) and total bytes (`800 KB`) occurred during a single execution of your function!

Best Practices:

- **Use Divan for Allocation Auditing:** When writing zero-allocation parsers or high-frequency networking code, run `cargo bench -- --alloc` in CI to ensure your functions maintain `0 B, 0 count` allocations!
- **Use Criterion when Statistical Regression Graphs are Needed:** If your team requires HTML regression charts and p-value proof across GitHub Pull Requests, use Criterion. If you want developer speed, simplicity, and allocation counting, use Divan!

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Guide: CPU Profiling with cargo-flamegraph & perf

Overview

`cargo-flamegraph` is a Rust-native wrapper around Linux's `perf` subsystem (and DTrace on macOS). It collects high-frequency CPU stack trace samples and generates interactive SVG FlameGraphs to visualize hot code paths and execution bottlenecks.

1. How to Download & Install (`download`)

Prerequisites (Linux)

You must install the Linux kernel performance counters tool (`perf`):

```
# Ubuntu / Debian
sudo apt update && sudo apt install -y linux-tools-common linux-tools-generic linux-tools-$(uname -r)

# Fedora / RHEL
sudo dnf install -y perf

# Arch Linux
sudo pacman -S perf
```

Allow Non-Root Profiling (Linux)

By default, Linux restricts `perf` to root users. To allow non-root users to record CPU events:

```
# Temporary (until reboot):
sudo sysctl -w kernel.perf_event_paranoid=-1

# Permanent:
echo "kernel.perf_event_paranoid = -1" | sudo tee -a /etc/sysctl.conf
```

Install `cargo-flamegraph`

Install the Rust CLI wrapper directly via Cargo:

```
cargo install cargo-flamegraph
```

2. How to Configure for Debugging (`debug`)

To generate readable FlameGraphs, the profiler **must** have access to debug symbols (function names and line numbers). If you compile in standard `--release` mode without symbols, your FlameGraph will just show `[unknown]` or hex memory addresses!

Add the following configuration to your project's `Cargo.toml`:

```
[profile.release]
debug = true      # Keep debug symbols in release builds (essential!)
strip = false     # Do not strip symbols
lto = "thin"      # Optional: Link-time optimization for realistic performance
```

3. How to Run & Profile (`profile`)

Profiling a Binary Application

To compile your app in release mode and immediately generate a FlameGraph:

```
cargo flamegraph --bin my_app -- --my-arg-1 --my-arg-2
```

- Note: Everything after `--` is passed directly as arguments to your application.

Profiling Unit Tests or Integration Tests

You can also profile specific test cases to find bottlenecks in algorithms:

```
cargo flamegraph --unit-test --test test_name -- --nocapture
```

Profiling a Running Process (PID)

If your Rust server or daemon is already running in the background:

```
sudo flamegraph --pid <PID> --output server-flamegraph.svg
```

4. Interpreting Results & Best Practices

When execution finishes, `cargo-flamegraph` creates an interactive SVG file named `flamegraph.svg` in your current directory. Open it in any web browser (Chrome, Firefox, Safari):

```
xdg-open flamegraph.svg # Linux
open flamegraph.svg    # macOS
```

How to Read the FlameGraph:

1. **Y-Axis (Stack Depth):** Represents the call stack hierarchy. The bottom box is `main()`, and boxes stacked on top are the functions called by their parents.
2. **X-Axis (Population / CPU Samples): This is NOT time!** The width of a box represents the **total percentage of CPU samples** that function occupied during the entire run.
3. **Plateaus (Wide Horizontal Boxes):** A wide box at the very top of a stack represents a **CPU Hotspot**—a function actively consuming massive amounts of processor time!
4. **Interactive Zoom:** Click on any box in the browser to zoom in on that specific call tree. Click "Reset Zoom" in the top left to zoom out.

Optimization Workflow:

- Identify the widest top-level boxes.
- Check if time is spent in memory allocation (`malloc` / `free`), hashing (`SipHash` vs `FxHash` / `AHash`), or cloning (`<T as Clone>::clone`).
- Optimize the hot algorithm, re-run `cargo flamegraph`, and verify that the plateau shrinks!

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Guide: Source-Based Code Coverage with `cargo-llvm-cov`

Overview

`cargo-llvm-cov` is the modern industry standard for tracking code coverage in Rust. It leverages LLVM's native source-based coverage instrumentation (built directly into the `rustc` compiler via `-C instrument-coverage`). It produces 100% exact line, branch, and region coverage without needing external debugging utilities or `ptrace`.

1. How to Download & Install (`download`)

Step 1: Install the Cargo Subcommand

Install `cargo-llvm-cov` via Cargo:

```
cargo install cargo-llvm-cov
```

Step 2: Install the LLVM Tools Preview Component

Because `llvm-cov` relies on LLVM binaries shipped with the Rust toolchain (`llvm-cov` and `llvm-profdata`), install the Rustup component:

```
rustup component add llvm-tools-preview
```

2. How to Configure for Debugging (`debug`)

`cargo-llvm-cov` works seamlessly with standard `cargo test` suites without requiring edits to `Cargo.toml`.

However, if you want to test code coverage across multi-threaded tests or integration tests that spawn subprocesses, set the following environment variable in your terminal or CI pipeline:

```
# Ensure subprocesses inherit coverage instrumentation tracking:  
export CARGO_LLVM_COV_SETUP=1
```

Enabling Branch Coverage (Nightly Rust Only)

By default, stable Rust tracks **Line Coverage** and **Region Coverage**. To track **Branch Coverage** (checking whether both `if` and `else` paths were evaluated), run with Rust Nightly:

```
rustup toolchain install nightly
```

3. How to Run & Profile Coverage (`profile`)

Generate a Quick Terminal Summary

To run your entire test suite and print a clean percentage table to the console:

```
cargo llvm-cov
```

Sample Output:

Filename Cover	Regions	Missed Regions	Cover	Lines	Missed Lines

src/lib.rs 96.67%	45	3	93.33%	120	4
src/parser.rs 92.90%	112	18	83.93%	310	22

TOTAL 93.95%	157	21	86.62%	430	26

Generate and Open an Interactive HTML Report

To generate a detailed line-by-line HTML report and automatically open it in your browser:

```
cargo llvm-cov --open
```

Generate LCOV / Cobertura Reports for CI/CD

For integration with GitHub Actions, Codecov, or SonarQube:

```
# Generate lcov.info:
cargo llvm-cov --lcov --output-path lcov.info

# Generate Cobertura XML:
cargo llvm-cov --cobertura --output-path cobertura.xml
```

4. Interpreting Results & Best Practices

When viewing the interactive HTML report (`target/llvm-cov/html/index.html`):

1. **Bright Green Lines:** Code paths that were executed at least once during `cargo test`. The number in the left margin shows the exact execution count (e.g., `42x`).
2. **Bright Red Lines:** Code paths that were **never executed** by any test case!
3. **Yellow / Partial Regions:** Lines containing multiple branches (e.g., `if a && b`) where only one condition was tested.

Best Practices for Systems Programmers:

- **Target Error Handling:** In systems programming, unhandled error paths (`Err(...)` or `match` fallback arms) are the #1 source of production bugs. Use `llvm-cov` to ensure every custom error enum variant is explicitly triggered by a test!
- **Ignore Boilerplate:** Use the `#[cfg(not(tarpaulin_include))]` or `#[no_coverage]` (nightly) attributes to exclude derived traits or CLI formatting boilerplate from skewing your coverage statistics.
- **CI Quality Gating:** Enforce a minimum coverage threshold in CI: `bash`
`cargo llvm-cov --fail-under-lines 85`

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Guide: Multi-Threaded CPU Profiling with Samplify & Firefox Profiler

Overview

`samplify` is a state-of-the-art CPU sampling profiler for macOS and Linux. It records application execution and automatically streams the telemetry into the world-class **Firefox Profiler** web application (`profiler.firefox.com`). It is especially renowned for its unmatched visualization of multi-threaded Rust work pools (like Tokio or Rayon) and seamless Apple Silicon (macOS) support.

1. How to Download & Install (`download`)

`samplify` is a standalone command-line tool installed via Cargo:

```
cargo install samplify
```

- **macOS Note:** No system kernel modifications or root privileges are required on macOS!
- **Linux Note:** Ensure `kernel.perf_event_paranoid` is set to `-1` or `1` (see Linux perf setup in `flamegraph.profile.md`).

2. How to Configure for Debugging (`debug`)

Just like all sampling profilers, `samplify` requires debug symbols to map memory addresses to Rust function names and filenames.

Add this to your project's `Cargo.toml`:

```
[profile.release]
debug = true      # Retain DWARF debug symbols
strip = false     # Do not strip binary symbols
```

Optional: Enabling Frame Pointers (For Extra Call Stack Accuracy)

On some x86_64 architectures, enabling frame pointers ensures 100% perfect stack unwinding without reliance on DWARF tables:

```
# Set in your terminal environment before building:
export RUSTFLAGS="-C force-frame-pointers=yes"
```

3. How to Run & Profile (`profile`)

Profiling a Cargo Project Directly

You can prefix any `cargo run` command with `sample record`:

```
sample record cargo run --release --bin my_app -- --arg1 --arg2
```

Profiling a Pre-compiled Binary

If you have already built your release binary:

```
sample record ./target/release/my_app
```

What Happens When Execution Ends?

1. `sample` compresses the recorded stack samples into a local profile profile.
2. It spins up a temporary local web server (e.g., on `http://127.0.0.1:3000`).
3. It **automatically opens your default web browser** to `https://profiler.firefox.com` and loads the profile directly from your local machine! * *Privacy Note*: All data is processed 100% locally inside your browser; no profiling data is ever uploaded to Mozilla or external servers!

4. Interpreting Results & Best Practices

The Firefox Profiler UI is arguably the most powerful open-source visualization suite in systems programming.

Key Features to Explore:

1. **Thread Timeline (Top Panel)**: Shows every OS thread spawned by your Rust app (e.g., `tokio-runtime-worker-0`, `rayon-core-1`). You can see exactly when threads are executing CPU work, sleeping, or blocked on locks!
2. **Call Tree (Bottom Left)**: A hierarchical breakdown of total execution time per function, sorted by **Self Time** (time spent directly in the function) versus **Total Time** (time spent in function + its children).
3. **Flame Graph Tab**: An interactive FlameGraph similar to `cargo-flamegraph`, but with instant search, filtering by thread, and time-slice selection.

4. **Stack Chart / Timeline Zoom:** Highlight any time range in the top timeline (e.g., during a latency spike or initialization drop) to filter the Call Tree and Flame Graph exclusively to that exact millisecond window!

Best Practices:

- Use `sample` when debugging async Tokio deadlocks or Rayon thread-pool starvation.
- Look for gaps in worker thread timelines—if 7 out of 8 threads are idle while 1 thread is at 100% CPU, your workload is poorly parallelized or suffering from lock contention!

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Guide: Real-Time Frame & Execution Timeline Profiling with Tracy

Overview

Tracy Profiler is a real-time, nanosecond-resolution frame and execution timeline profiler. Widely used in systems programming, game engines (such as Bevy), robotics, and real-time audio/video processing, Tracy streams live performance telemetry from your running Rust application directly to the graphical Tracy desktop client, visualizing thread contention, lock delays, frame rate spikes, and memory allocations over time.

1. How to Download & Install (`download`)

Tracy consists of two parts: the **Desktop Client GUI** (the viewer) and the **Rust Library Crates** (the instrumentation).

Step 1: Download the Tracy Desktop Client GUI

Download the pre-compiled Tracy Profiler graphical client for your OS (Windows, macOS, or Linux) from the official GitHub releases: * <https://github.com/wolfpld/tracy/releases>

Step 2: Add Rust Dependencies

Hook Tracy directly into Rust's standard `tracing` ecosystem by adding `tracing` and `tracing-tracy` to your `Cargo.toml`:

```
[dependencies]
tracing = "0.1"
tracing-subscriber = "0.3"
tracing-tracy = "0.11"
```

2. How to Configure for Debugging (`debug`)

In your application's entry point (`src/main.rs`), initialize the `tracing-subscriber` with the `TracyLayer`. Then, annotate any function or async task you want to track on the timeline with `#[tracing::instrument]`.

Runnable Instrumentation Template (`src/main.rs`):

```
use std::time::Duration;
use tracing::{info, instrument};
use tracing_subscriber::layer::SubscriberExt;
use tracing_subscriber::util::SubscriberInitExt;

// 1. Annotate functions to automatically emit begin/end timeline spans!
#[instrument]
fn perform_heavy_database_query(query_id: u32) {
    info!("Executing query {}", query_id);
    std::thread::sleep(Duration::from_millis(15)); // Simulate CPU/IO work
}

#[instrument]
fn render_game_frame(frame_num: u64) {
    perform_heavy_database_query(1);
    perform_heavy_database_query(2);
    std::thread::sleep(Duration::from_millis(5));
}

fn main() {
    // 2. Initialize the Tracy layer to stream telemetry on port 8042
    tracing_subscriber::registry()
        .with(tracing_tracy::TracyLayer::default())
        .init();

    println!("Connecting to Tracy Profiler... (Open Tracy GUI now!)");

    // 3. Main application loop (simulating 60 FPS frame updates)
    for frame in 1..=500 {
        render_game_frame(frame);

        // Mark the end of a graphical or logical frame in Tracy:
        tracing_tracy::client::frame_mark();
    }
}
```

3. How to Run & Profile (`profile`)

Step 1: Launch the Tracy Desktop GUI

Open the downloaded `Tracy` application on your desktop. You will see a button labeled **Connect (localhost)**.

Step 2: Run Your Rust Application

Launch your application in release mode:

```
cargo run --release
```

Step 3: Connect Live in the GUI

As soon as your Rust app starts, click **"Connect"** in the Tracy GUI! You will see live, real-time telemetry streaming across your screen at 60+ FPS!

4. Interpreting Results & Best Practices

The Tracy GUI provides deep insight into real-time system behavior:

Key Features in the Tracy GUI:

1. **Frame Time Graph (Top Bar):** Shows a bar chart of every frame's duration in milliseconds. Spikes above `16.6 ms` indicate dropped frames (falling below 60 FPS)! Click on any spike to inspect that exact frame!
2. **Timeline Tracks (Center):** Displays horizontal tracks for every OS thread and async worker. You will see nested colored boxes representing your `#[instrument]` function spans (`render_game_frame` → `perform_heavy_database_query`).
3. **Lock Contention Tracking:** Tracy automatically highlights Mutex/RwLock waiting times in red, showing you exactly which thread was holding the lock and causing another thread to stall!
4. **Memory Profiling Tab:** If configured with Tracy's global allocator wrapper, displays live graphs of heap memory fragmentation and total allocated blocks over time.

Best Practices:

- **Feature Gating in Production:** Keep Tracy instrumentation zero-cost in production by wrapping the subscriber initialization behind a Cargo feature flag: `toml`
`[features]`
`profiling = ["tracing-tracy"]` Run with `cargo run --release --features profiling` only when live profiling is needed!
- **Async Tokio Tracking:** Tracy works seamlessly with async/await! When using Tokio, `#[tracing::instrument]` correctly tracks async futures across worker thread migrations!

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Guide: Deterministic CPU & Cache Miss Debugging with Valgrind Callgrind

Overview

Valgrind Callgrind is a CPU emulation and profiling tool that records deterministic execution metrics. Instead of sampling wall-clock time (which fluctuates due to OS background tasks or CPU boost clock scaling), Callgrind simulates a CPU to count exact machine instructions executed (`Ir`), L1/L2/L3 cache misses, and conditional branch mispredictions. It is paired with **KCachegrind** (or QKachegrind) for visual call-graph analysis.

1. How to Download & Install (`download`)

Linux Installation

Install Valgrind and the KCachegrind GUI via your package manager:

```
# Ubuntu / Debian
sudo apt update && sudo apt install -y valgrind kcachegrind

# Fedora / RHEL
sudo dnf install -y valgrind kcachegrind

# Arch Linux
sudo pacman -S valgrind kcachegrind
```

macOS Installation

On Apple Silicon macOS, native Valgrind support is limited. We recommend running Callgrind inside a Docker Linux container:

```
docker run --rm -it -v $(pwd):/workspace -w /workspace rust:latest bash
# Inside container: apt update && apt install -y valgrind
```

- Note: For native macOS visualization, install `qcachegrind` via Homebrew: `brew install qcachegrind`.

2. How to Configure for Debugging (`debug`)

To map machine instructions and cache misses directly to your Rust source code lines, ensure DWARF debug symbols are enabled without stripping.

Add this configuration to `Cargo.toml` :

```
[profile.release]
debug = true      # Required to map instructions to Rust file line numbers!
strip = false
lto = "thin"
```

Isolating Specific Function Boundaries

When analyzing small algorithms, LLVM function inlining can merge child functions into their parents, obscuring call graphs. To forbid inlining on a function being investigated, use the attribute:

```
#[inline(never)]
pub fn critical_parsing_routine(data: &[u8]) -> u32 {
    // Hot algorithm here...
}
```

3. How to Run & Profile (`profile`)

Step 1: Compile Your Release Binary

```
cargo build --release
```

Step 2: Execute Under Callgrind Emulation

Run your compiled binary under Valgrind with instruction dumping and jump/branch tracking enabled:

```
valgrind --tool=callgrind \
  --dump-instr=yes \
  --collect-jumps=yes \
  --simulate-cache=yes \
  ./target/release/my_app --arg1 --arg2
```

- `--simulate-cache=yes` : Simulates L1 and LL (Last Level / L3) CPU data and instruction caches to count exact cache misses!
- `--collect-jumps=yes` : Tracks conditional branch predictions and mispredictions.

What Happens When Execution Ends?

Valgrind writes a deterministic profile file named `callgrind.out.<PID>` in your current working directory.

4. Interpreting Results & Best Practices

Open the generated profile file using **KCachegrind** (Linux) or **QCachegrind** (macOS/Windows):

```
kcachegrind callgrind.out.<PID>
```

Key Metrics in KCachegrind:

1. **Ir** (**Instruction Read / Executed**): The total number of x86_64 or ARM assembly instructions executed. **This number is 100% deterministic!** Running the exact same input on a supercomputer or a Raspberry Pi will yield the exact same **Ir** count!
2. **I1mr** / **ILmr** (**Instruction Cache Misses**): Code too large to fit in L1/L3 instruction caches, causing CPU pipeline stalls.
3. **D1mr** / **DLmr** (**Data Cache Misses**): The #1 killer of systems performance! Indicates your algorithm is jumping across fragmented RAM (e.g., pointer-chasing through linked lists or scattered **Box<T>** nodes) instead of accessing contiguous memory (like **Vec<T>**).
4. **Bc** / **Bcm** (**Branch Conditional Mispredictions**): Unpredictable **if** / **else** branches that flush the CPU speculative execution pipeline.

Systems Optimization Workflow:

- Replace pointer-based data structures (**LinkedList** , graph nodes with **Rc<RefCell<T>>**) with contiguous flat arrays (**Vec<T>** , **SlotMap**).
- Re-run Callgrind and watch **D1mr** (Data Cache Misses) plummet by orders of magnitude!

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